ChurnGuide: 25 Viva Questions and Answers

Prepared for the B.Tech project viva. Based on the actual code and results of the submitted project.

Twenty-five practical questions an examiner may ask, with short answers based on the actual project code and saved results. Learn the ideas, not the exact words. Where a limitation is asked, answer it honestly.

Q1. What is the aim of your project?

To build a local web app that predicts how likely each customer is to churn, explains the prediction with model sensitivity, scores a whole customer file, and groups customers into segments. It is built to accept different business CSV files, not only one dataset.

Q2. What is customer churn?

Churn means a customer stops or cancels the service. In the telecom data a customer who left has Churn = Yes. In our data 26.5% (1,869 of 7,043) churned.

Q3. Which datasets did you use?

The public IBM Telco Customer Churn sample (7,043 customers) for the main results, and a synthetic B2B SaaS dataset of 1,200 accounts generated by our own script (scripts/generate_saas_data.py) to test that the app handles different columns.

Q4. Is the SaaS data real?

No. It is synthetic. Its results (accuracy 0.6542, ROC-AUC 0.6894) show the software works with another column set. They are not evidence about real SaaS companies.

Q5. Which models did you compare and why?

Logistic Regression (simple baseline), Random Forest and Gradient Boosting (both can learn interactions between features). They give different bias-variance trade-offs, and we let cross-validation choose.

Q6. How was the best model chosen?

Using stratified 5-fold cross-validation on the 80% training part only, scored by ROC-AUC, with a small hyperparameter grid for each model. Gradient Boosting had the highest mean (0.8493), then Logistic Regression (0.8463) and Random Forest (0.8446). The differences are small, with standard deviations around 0.012.

Q7. What are the final test results?

On 1,409 held-out telecom customers: accuracy 0.7984, ROC-AUC 0.8465, precision 0.6585, recall 0.50, F1 0.5684. Confusion matrix: 938 true negatives, 97 false positives, 187 false negatives, 187 true positives.

Q8. Is 0.8493 the accuracy?

No. 0.8493 is the cross-validated ROC-AUC of the winner on the training data. The home page card labels it 'Top Accuracy', which is a labelling mistake in the interface. The held-out accuracy is 0.7984.

Q9. What does ROC-AUC mean?

It is the chance that the model gives a higher risk score to a randomly chosen churner than to a randomly chosen stayer. 0.5 is guessing and 1.0 is perfect. Ours is 0.8465 on the test set.

Q10. Why is accuracy alone not enough here?

Only 26.5% of customers churn, so a model that says nobody churns would be about 73.5% accurate. That is why we also report recall, precision, F1 and ROC-AUC.

Q11. What is recall and what is yours?

Recall is the share of real churners the model catches. Ours is 0.50, so it misses half of them (187 of 374). This is a real weakness.

Q12. What is precision and what is yours?

Precision is the share of flagged customers who really churn. Ours is 0.6585, so about two in three flagged customers are real churners and one in three is a false alarm (97 false positives against 187 true positives).

Q13. Could you improve recall?

Possibly, by lowering the decision cut-off or using cost-aware tuning, but that would also raise false alarms. We have not tested this, so we list it as future work that needs business validation.

Q14. How did you avoid data leakage?

The split is made first (80/20 stratified, seed 42). Imputation, scaling and encoding sit inside one scikit-learn Pipeline, so they are fitted on training data only. Model selection uses only training cross-validation. The test set is used once for the final report. The wizard also asks the user to confirm that all columns were known before the churn outcome.

Q15. Does that prove there is no leakage?

No. These steps remove the common technical leaks. They cannot prove that a user's columns were truly known before the outcome. That needs the user's judgement about the data.

Q16. How are categorical and numeric columns handled?

Numeric columns: median imputation and standard scaling. Categorical columns: most-frequent imputation and one-hot encoding (first level dropped, unknown levels ignored).

Q17. How does the app explain one prediction?

It changes one input at a time to a typical training value (median or mode) and records how the churn probability changes. That is model sensitivity. It is not SHAP and it is not causal. Missing baselines are filled with saved typical values, and TotalCharges is recalculated when tenure or charges change.

Q18. What are permutation feature importances?

On the test set, one column at a time is shuffled and the drop in ROC-AUC is measured (10 repeats). A bigger drop means the model relies on that column more. For telecom the top ones are tenure, then Contract and InternetService.

Q19. Does a top feature mean it causes churn?

No. It shows what the model relies on, not what causes churn. The data is observational.

Q20. How does segmentation work?

K-Means clustering (k = 4) on standardized numeric columns: tenure, MonthlyCharges and TotalCharges. Historical churn in the four telecom segments is 48.2%, 24.6%, 15.4% and 5.0%. The churn label is not used to build the clusters.

Q21. What are the risk cut-offs?

A probability of 0.60 or more is high risk for single predictions and the simulator. Bulk scoring uses 0.65 for high risk. They are fixed values in the code, not tuned for a business.

Q22. How does bulk scoring work?

The user uploads a CSV without a churn column. The saved pipeline gives each row a probability and risk level, the page shows a preview, and the scored file can be downloaded. The scored file is one shared file, so score again after switching business.

Q23. How does the app handle a different business?

The three-step wizard infers column types, asks the user to confirm the target column and the meaning of churn, checks the target is binary with enough examples per class, and trains an isolated project. Automatic guesses can be wrong, so the mapping must be checked for any new CSV.

Q24. What are the main limitations?

One snapshot per customer (no time series), recall of 0.50, synthetic second dataset, sensitivity explanations that are not causal, fixed cut-offs, a mobile layout problem on the home page, and untested Windows/Mac launchers.

Q25. How did you test the software, and what failed?

Unit tests for data loading, pipeline and leakage checks, explanations and web routes. With the pinned Python 3.11, 17 passed and 1 failed. The failing test is an old navigation test that expects a redirect, but the home page now loads directly. We did not change the code.

Quick facts to remember

Item	Value
Telecom dataset	7,043 customers, 26.5% churn
Split	80/20 stratified, seed 42 (5,634 train, 1,409 test)
Winner	Gradient Boosting, CV ROC-AUC 0.8493
Test accuracy / AUC	0.7984 / 0.8465
Recall / precision	0.50 / 0.6585
SaaS (synthetic)	1,200 accounts, Logistic Regression, AUC 0.6894
Segments	K-Means, k = 4, churn 48.2 / 24.6 / 15.4 / 5.0 %